

OPEN Compute Project

HPM and Peripheral Programmable Device (PD) Specification: Schema Documentation

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L = Local, **R** = Required

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Top-level Properties

Description: The Programmable Device FRU schema defines the hardware and firmware properties of a field programmable gate array or equivalent programmable device used to enable a management interface for an HPM.

Property	Type	L	R	Description
_comment	string			A general comment about a data file
Copyright	string			A copyright statement for the FRU data.
Version	string	Y	Y	The current file version for this FRU data.
SchemaVersion	string	Y	Y	The revision of the Programmable Device schema that this FRU data was built against.
GitHubChecksum	string			The GitHub checksum for this FRU data.
Author	string			The author of this FRU data.
General	General	Y	Y	The General section contains general information about the Programmable Device plus overall build information.
IPBlocks	IPBlocks	Y	Y	Information about the IP blocks compiled into this Programmable Device.
MemoryMap	MemoryMap	Y	Y	Information about the memory map for this Programmable Device.
Registers	Registers			Information defining the registers exported by this Programmable Device.
PhysicalSignals	PhysicalSignals	Y	Y	Information about the physical signals supported by this Programmable

Property	Type	L	R	Description
				Device.
VirtualWires	VirtualWires	Y	Y	Information about the virtual wires supported by this Programmable Device.
Buses	Buses	Y	Y	Information about the logical and physical buses connected to this Programmable Device.

General

Description: The General section contains general information about the Programmable Device plus overall build information.

Property	Type	L	R	Description
VendorID	string	Y	Y	The vendor ID for this Programmable Device.
UniqueID	string	Y	Y	The unique ID specific to this Programmable Device.
Model	string			The model for this Programmable Device.
Manufacturer	string			The manufacturer for this Programmable Device.
Build	General.Build			Information about the firmware build for this Programmable Device.

General.Build

Description: Information about the firmware build for this Programmable Device.

Property	Type	L	R	Description
Version	string			The version number for the Programmable Device build.
BuildTimestamp	string			The timestamp when the Programmable Device build completed.
Builder	string			The name of individual who performed the build.

IPBlocks

Description: Information about the IP blocks compiled into this Programmable Device.

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Type	string	Y	Y	The type of this IP block.
Spec	SpecReference			A reference to an OCP specification.
GUID	string	Y	Y	A globally unique identifier for this IP block.
HWID	string	Y	Y	A hardware identifier for this IP block.
SupportedSOCSockets	integer			The maximum number of CPU SOC sockets supported by this IP block.
ExtendedVersion	IPBlocks.ExtendedVersion			Extended version information for this IP block.
ConnectedBuses	ReferencedBusList			A collection of references to hardware or logical buses.

Property	Type	L	R	Description
OEM	object			OEM-specific information for this IP Block.

IPBlocks.ExtendedVersion

Description: Extended version information for this IP block.

Property	Type	L	R	Description
Originator	string			The originating entity for this IP block.
OriginalIPVersion	string			The source version for this IP block, prior to modifications.
ModifiedBy	string			The name of the entity that has modified this IP block.
ModificationVersion	string			The modification version for this IP block.
ModificationTimestamp	string			The timestamp for when modifications to this IP block were completed.
ModificationContact	string			Contact information for the author of modifications to this IP block, such as an email address.

MemoryMap

Description: Information about the memory map for this Programmable Device.

Property	Type	L	R	Description
Version	string			The current version for this MemoryMap data.
Regions	array of MemoryMap.Regions	Y	Y	Information about different regions of memory in the map.

MemoryMap.Regions

Description: Information about different regions of memory in the map.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
MappedIPBlock	string	Y	Y	The IP Block to which this memory region maps.
Version	string			The version of this IP block for CSRs.
RegionOffsetBytesHex	string	Y	Y	The offset in memory for the beginning of this region, in hexadecimal format.
RegionSizeBytes	integer	Y	Y	The size in bytes for this memory region.
RegisterMap	array of MemoryMap.Regions.RegisterMap	Y	Y	Information about registers in the map.
BAR	array of MemoryMap.Regions.BAR			Information about the Base Address Registers (BARs) of this device.

MemoryMap.Regions.RegisterMap

Description: Information about registers in the map.

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.

Property	Type	L	R	Description
Spec	SpecReference	Y	Y	A reference to an OCP specification.
XIOs	integer			The number of XIOs that the register set represents.
PICPWRs	integer			The number of PICPWRs that the register set represents.
OffsetBytes	integer	Y	Y	The offset from the beginning of this region for this set of registers.
SizeBytes	integer	Y	Y	The size in bytes for this set of registers.
MappedRegister	string			The named register to which this register maps.
InactiveRanges	array of MemoryMap.Regions.RegisterMap.InactiveRanges			Ranges of registers from the spec not active.
AccessDomain	string			Which agent(s) may access this register/register-set. Possible values: `BMCOnly`, `HostOnly`, `Shared`
HostAccessDescription	string			Description of host-visible behavior, required when AccessDomain is HostOnly or Shared.

MemoryMap.Regions.RegisterMap.InactiveRanges

Description: Ranges of registers from the spec not active.

Property	Type	L	R	Description
Start	integer			No description available
Width	integer			No description available

MemoryMap.Regions.BAR

Description: Information about the Base Address Registers (BARs) of this device.

Property	Type	L	R	Description
Index	integer			Base Address Register index.
BaseAddressHex	string			Base address value advertised to the host, in hexadecimal.
SizeBytes	integer			Size of the BAR window.
Prefetchable	boolean			Whether the BAR is marked prefetchable.
AddressWidth	string			32-bit or 64-bit BAR. Possible values: `32` , `64`

Registers

Description: Information defining the registers exported by this Programmable Device.

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
SizeBytes	integer			The size in bytes of this register.
Bitfields	array of Registers.Bitfields			The bitfields defined within this register.

Registers.Bitfields

Description: The bitfields defined within this register.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Type	string			The type of this bitfield element.
LeastSignificantBit	integer			The least significant bit for this bitfield element.
SizeBits	integer			The size in bits of this bitfield element.
Reset	integer			The value of this bitfield element after a reset.
Access	string			The type of access intended for this bitfield element. Possible values: `ro` (read-only access: the bitfield element may be read but should not be written to), `rw` (read/write access: the bitfield element may be read or written to), `wo` (write-only: the bitfield element may be written to but should not be read), `w1c` (write one to clear: writing a 1 to a bit in the bitfield element clears that bit.)
EnumeratedValues	array of Registers.Bitfields.EnumeratedValues			The enumerated values for this bitfield element, if applicable.

Registers.Bitfields.EnumeratedValues

Description: The enumerated values for this bitfield element, if applicable.

Property	Type	L	R	Description
Value	integer			The numeric value of the enumerated option.
Name	string			The symbolic name of the enumerated option.
Description	string			A vendor-defined description of the state/code.
VendorDefined	boolean			Indicates if this value falls within an OEM/vendor-reserved range.

PhysicalSignals

Description: Information about the physical signals supported by this Programmable Device.

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
IdentifierPrefix	string			The identifier prefix for a set of physical signals.
FirstIdentifier	integer			The first number the identifier will iterate from for the set of physical signals.
PhysicalPin	string			The package pin for this signal.
PhysicalPinPrefix	string			The physical pin prefix for a set of physical signal pins.
FirstPhysicalPin	integer			The first physical pin number for a set of physical signal pins.

Property	Type	L	R	Description
PhysicalPinCount	integer			The number of physical pins in a set of physical signal pins.
Direction	Direction			The direction of a signal, virtual wire, or GPIO.
DriveCapabilities	array of string			Output drive capabilities for this signal when configured as an output.
PowerWell	RequiredPowerState			The power domain of the voltage rail needed for this signal.
PullUp	boolean			Indicates that this signal has a pull-up resistor.
PullDown	boolean			Indicates that this signal has a pull-down resistor.
IOType	string			The I/O standard for this signal. Possible values: `LVCMOS12`, `LVCMOS15`, `LVCMOS18`, `LVCMOS25`, `LVCMOS33`

VirtualWires

Description: Information about the virtual wires supported by this Programmable Device.

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
Start	integer	Y	Y	The first virtual wire in the set.
TotalWidth	integer	Y	Y	The total number of virtual wires in the set.

Property	Type	L	R	Description
Behavior	VirtualWires.Behavior			Definition for a set of virtual wires.
InternalNetlistMappingNames	array of NameSet			Internal netlist names used to map a set of virtual wires.
BusMappings	array of VirtualWires.BusMappings			Information about buses mapped to virtual wires.
RegisterMappings	array of VirtualWires.RegisterMappings			Information about registers mapped to virtual wires.
PhysicalSignalMappings	array of VirtualWires.PhysicalSignalMappings			Information about physical signals mapped to virtual wires.
IPBlockMappings	array of VirtualWires.IPBlockMappings			Information about IP blocks mapped to virtual wires.
Capabilities	array of VirtualWires.Capabilities			Capabilities that control the behavior of virtual wires.

VirtualWires.Behavior

Description: Definition for a set of virtual wires.

Property	Type	L	R	Description
DefaultState	integer	Y	Y	The default state for virtual wires in the set.
ActiveState	integer	Y	Y	The active state for virtual wires in the set.

Property	Type	L	R	Description
Direction	Direction	Y	Y	The direction of a signal, virtual wire, or GPIO.
StartupPolarity	string	Y	Y	The initial polarity for virtual wires in the set. Possible values: `High` (A logical high means the signal is in an active state), `Low` (A logical low means the signal is in an active state), `HighImpedance` (The signal is in a logically disabled state)
RequiredPowerState	RequiredPowerState			The minimum power state to utilize a bus.

VirtualWires.BusMappings

Description: Information about buses mapped to virtual wires.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
Interfaces	array of string			The specific buses that virtual wires are being mapped to.
Start	integer			The index of the first virtual wire being mapped to a bus.
Width	integer			The number of virtual wires being mapped to buses.
NameSet	NameSet			The names of buses that the virtual wires are being mapped to.
Signals	array of string			The set of signals that virtual wires are being mapped to virtual wires.
Description	string			A description of the mapping.

VirtualWires.RegisterMappings

Description: Information about registers mapped to virtual wires.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
Start	integer			The index of the first virtual wire being mapped to a register.
Width	integer			The number of virtual wires being mapped to registers.
NameSet	NameSet			The names of registers that the virtual wires are being mapped to.
RegionNames	array of string			The regions containing the registers that virtual wires are being mapped to.
ManyToOne	boolean			Indicates that a mapping applies to all virtual wires in the `InternalNetlistMappingNames` set.
Bitfields	array of string			A list of named bitfields in the register that the virtual wires are mapped to.
Description	string			A description of the mapping.

VirtualWires.PhysicalSignalMappings

Description: Information about physical signals mapped to virtual wires.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
Start	integer			The index of the first virtual wire being mapped to a physical signal.

Property	Type	L	R	Description
Width	integer			The number of virtual wires being mapped to physical signals.
NameSet	NameSet			The names of physical signals that the virtual wires are being mapped to.
Description	string			A description of the mapping.

VirtualWires.IPBlockMappings

Description: Information about IP blocks mapped to virtual wires.

Property	Type	L	R	Description
Alias	ComponentAlias			A secondary reference name for a component.
Start	integer			The index of the first virtual wire being mapped to an IP block.
Width	integer			The number of virtual wires being mapped to IP blocks.
ManyToOne	boolean			Indicates that a mapping applies to all virtual wires in the `InternalNetlistMappingNames` set.
NameSet	NameSet			The names of IP blocks that the virtual wires are being mapped to.
Description	string			A description of the mapping.

VirtualWires.Capabilities

Description: Capabilities that control the behavior of virtual wires.

Property	Type	L	R	Description
NameSet	NameSet			The virtual wires to which the capability flags apply.

Property	Type	L	R	Description
Bitmask	integer			The capabilities bitmask for a virtual wire.
MaskType	string			The type of bitmask that applies to this virtual wire. Possible values: `FlexIO`, `Sideband`, `OEM`

Buses

Description: Information about the logical and physical buses connected to this Programmable Device.

Array Item Properties:

Property	Type	L	R	Description
BusType	string	Y	Y	The type for this bus. Possible values: `DisplayPort`, `ESPI`, `I2C`, `I3C`, `JTAG`, `LTPI`, `MPESTI`, `NCSI_RBT`, `PCIe`, `PECI`, `QSPI`, `SGMII`, `SPI`, `SGPIO`, `UART`, `USB`, `OEM`
Identifier	NamedComponent	Y	Y	A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
RequiredPowerState	RequiredPowerState			The minimum power state to utilize a bus.
SPDM	boolean			Indicates whether an SPDM connection can be established over this bus.
RandomAccessCapable	boolean			Indicates whether this bus supports random access into the memory space of the Programmable Device.

Property	Type	L	R	Description
HostAccessible	boolean			Indicates whether this bus is accessible to the host.
MaxClockSpeedMHz	number			Indicates the maximum clock speed this bus supports.
DataRateMultiplier	integer			Indicates a multiplication factor for the data rate for a bus, such as 2 for a double data rate (DDR) bus.
AlertSignal	string			Indicates the alert signal associated with this bus.
Tunneled	Buses.Tunneled			Information about how a bus is tunneled.
IPBlockMappings	array of string			Indicates the IP block that implements support for this bus.
Signals	array of Buses.Signals			Information about the signals associated with a bus.
Tunneling	Buses.Tunneling			Provides tunneling information for the bus.
MPESTI	Buses.MPESTI			Provides MPESTI-specific information for the bus.
SGPIO	Buses.SGPIO			Provides SGPIO-specific information for the bus.
ESPI	Buses.ESPI			Provides ESPI-specific information for the bus.
SPI	Buses.SPI			Provides SPI-specific information for the bus.
QSPI	Buses.QSPI			Provides QSPI-specific information for the bus.
OEM	Buses.OEM			Provides OEM-specific information for the bus.

Buses.Tunneled

Description: Information about how a bus is tunneled.

Property	Type	L	R	Description
Type	string			The type of bus over which this bus is tunneled. Possible values: `LTPI`, `OEM`
Interface	string			The specific interface over which this bus is tunneled.

Buses.Signals

Description: Information about the signals associated with a bus.

Property	Type	L	R	Description
PhysicalPin	string			Indicates the physical pin associated with this signal.
Alias	string			An alias for this signal.
NetName	string			The net name for this signal.

Buses.Tunneling

Description: Provides tunneling information for the bus.

Property	Type	L	R	Description
GPIOs	GPIOs			Information about a set of GPIOs.
TunneledBuses	array of Buses.Tunneling.TunneledBuses			Enumeration of buses tunneled through this LTPI bus.
DataChannelMappedRegions	array of Buses.Tunneling.DataChannelMappedRegions			Regions of memory mapped in the data channel.

Buses.Tunneling.TunneledBuses

Description: Enumeration of buses tunneled through this LTPI bus.

Property	Type	L	R	Description
BusType	string			The type of the tunneled bus.
BusId	string			Identifies a tunneled bus.

Buses.Tunneling.DataChannelMappedRegions

Description: Regions of memory mapped in the data channel.

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.
OffsetBytes	integer			The offset in memory to the beginning of this mapped memory region.
OffsetBytesHex	string			The offset in memory to the beginning of this mapped memory region, in hexadecimal.
SizeBytes	integer			The size in bytes for this mapped memory region.
SizeBytesHex	string			The size in bytes for this mapped memory region, in hexadecimal.
MappedMemoryRangeStartHex	string			The offset in the Programmable Device memory map for the beginning of the data channel mapped memory region, in hexadecimal.
MappedMemoryRegion	string			Indicates a memory region that is mapped to the data channel.

Property	Type	L	R	Description
Description	string			Describes a memory region that is mapped to the data channel.

Buses.MPESTI

Description: Provides MPESTI-specific information for the bus.

Property	Type	L	R	Description
ExternalMUXSupported	boolean			Indicates whether this MPESTI bus supports an external mux.
MaximumTargetsSupported	integer			Indicates the maximum number of targets supported by this MPESTI bus.
ConnectedToTarget	boolean			Indicates whether this bus is connected to an MPESTI target.
PayloadBufferDestructive	boolean			Indicates whether the discovery payload mailbox is a single destructive flat buffer holding only the most recently discovered target.
InitiatorsPerIPBlock	integer			Number of MPESTI initiators (PESTI wires) instances implemented within this IP block.
InitiatorBlockSetIdentifier	string			Identifier grouping this IP block into an Initiator IP Block Set.
InitiatorBlockSetCount	integer			Total number of Initiator IP Block Sets (Initiator Super Set size) supported by the Programmable Device.
OperationalMode	string			The operational mode of the M-PESTI MUX/Switch. Possible values: `Target`, `Snoop`
FanoutMethod	string			The method used to select/switch among M-PESTI channels. Possible

Property	Type	L	R	Description
				values: `MUX`, `Switch`

Buses.SGPIO

Description: Provides SGPIO-specific information for the bus.

Property	Type	L	R	Description
GPIOs	GPIOs	Y	Y	Information about a set of GPIOs.

Buses.ESPI

Description: Provides ESPI-specific information for the bus.

Property	Type	L	R	Description
DummyBytes	integer			Specifies the number of dummy padding bytes needed for the ESPI interface.

Buses.SPI

Description: Provides SPI-specific information for the bus.

Property	Type	L	R	Description
DummyBytes	integer			Specifies the number of dummy padding bytes needed for the SPI interface.

Buses.QSPI

Description: Provides QSPI-specific information for the bus.

Property	Type	L	R	Description
DummyBytes	integer			Specifies the number of dummy padding bytes needed for the QSPI

Property	Type	L	R	Description
				interface.

Buses.OEM

Description: Provides OEM-specific information for the bus.

No properties defined.

Common Definitions

ComponentAlias

Description: A secondary reference name for a component.

Type: string

Direction

Description: The direction of a signal, virtual wire, or GPIO.

Type: string

Allowed Values: "input", "output", "bidir"

GPIOs

Description: Information about a set of GPIOs.

Array Item Properties:

Property	Type	L	R	Description
Identifier	NamedComponent			A component that is named so as to be referenceable.
Alias	ComponentAlias			A secondary reference name for a component.

Property	Type	L	R	Description
Latency	string			The latency for this GPIO. Possible values: `Low`, `Normal`
StartBit	integer			The start bit for this GPIO.
WidthBits	integer			The width in bits for this GPIO.
VirtualWireEndpoint	string			Indicates the virtual wire attached to this GPIO.
Description	string			Describes the intended use for this GPIO.
Direction	Direction			The direction of a signal, virtual wire, or GPIO.
ByteOffset	integer			The byte offset for this GPIO.

NameSet

Description: An ordered set of names

Property	Type	L	R	Description
Count	integer			The number of names in the set
List	array of string			An explicit list of names
Block	NameSet.Block			A block definition of a set of names

NameSet.Block

Description: A block definition of a set of names

Property	Type	L	R	Description
NamePrefix	string			The name prefix for a set of names.
NameOffset	integer			An offset within a numerical suffix for a set of names.

NamedComponent

Description: A component that is named so as to be referenceable.

Type: string

ReferencedBusList

Description: A collection of references to hardware or logical buses.

Array Item Properties:

Property	Type	L	R	Description
Type	string			The type of bus for this reference.
Identifier	string			The instance of the bus type for this reference.
SpecBusName	string			The name used in the spec to reference this bus.

RequiredPowerState

Description: The minimum power state to utilize a bus.

Type: string

Allowed Values: "G3", "Pre-STBY", "STBY", "S5", "S0", "Other"

SpecReference

Description: A reference to an OCP specification.

Property	Type	L	R	Description
Name	string			The name of the OCP specification being referenced.
Version	string			The version of the OCP specification being referenced.

Property	Type	L	R	Description
Revision	string			The revision of the OCP specification being referenced.